ANDRÉ ROGÉRIO DA SILVA FILHO

Software Consultant & AI Engineer | Full-Stack · Autonomous Agents · Applied AI & Research

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SUMMARY

Software Consultant, AI Engineer, and Systems Researcher specializing in full-stack architecture, autonomous AI agents, and cyber-physical systems. Proven track record designing and delivering end-to-end production platforms using TypeScript, Python, React, Next.js, Node.js, and PostgreSQL. At CBI, acts as software consultant and solutions architect leading enterprise software engineering, autonomous AI agent architecture (Google Gemini, MCP servers, advanced RAG), cloud infrastructure on GCP (Cloud Run, Cloud SQL, Secret Manager), and database systems managing sensitive citizen data for metropolitan concessions. In research and residency (UniFacens/MCTI/Softex), engineers industrial Digital Twins over Private 5G networks and cognitive memory architectures with 3D spatial interfaces (Memory Lens), targeting international conference submission.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, React 19/18, Next.js 15, Express, Fastify, PyTest, Three.js, React Three Fiber, WebGL/GLSL Shaders

Applied AI & Agents: Autonomous AI Agents (LangGraph, MCP Servers), LLMs (Google Gemini API), Vision-Language Models (Qwen2-VL via vLLM on AMD ROCm), PyTorch, Advanced RAG (Hybrid Search & Reranking), NetworkX Relational Graphs, Structured Outputs (JSON Schema), Zod

Cloud, DevOps & Data: Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager), Docker, CI/CD Pipelines, PostgreSQL (Row-Level Security), Redis, Supabase, Prisma ORM, Git

Enterprise, Security & Compliance: LGPD/GDPR Compliance, AES-256-GCM Encryption, IDOR Protection, Microsoft 365, Entra ID, Microsoft Graph API, Microsoft Intune, Certified AppSec Practitioner (CAP)

PROJECTS

Cribum (AuditRx) | Multimodal Medical Audit & Fraud Graph

GitHub

AMD Developer Hackathon (Health Track) • Python, FastAPI, React 19, PyTorch, AMD ROCm, vLLM, Qwen2-VL, NetworkX, TypeScript

- Architected a local, privacy-first multi-agent medical claim and prescription audit system built for the AMD Developer Hackathon.
- Integrated local Open Vision-Language Models (Qwen2-VL served via vLLM on AMD ROCm) with uncertainty-aware multi-hypothesis extraction and clinical LASA safety databases.
- Engineered a relational multigraph anomaly detection engine using NetworkX, identifying systemic medical billing collusion and synthetic prescription fraud rings.

Memory Lens | Cognitive Computing Platform & 3D Neural Interface

GitHub

TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Gemini API, Vitest

- Designed and built a cognitive memory platform modeling human memory retention, associative recall, and structured forgetting (Ebbinghaus curve), reducing token storage overhead by up to 80%.
- Engineered a 3D bioluminescent neural organism ("Chief Organism") in React Three Fiber with custom GLSL shaders synchronized to system state via an event bus.
- Conducted as formal academic research on semantic document organization under Prof. Dr. Fabrício Londero (UniFacens), targeting submission to the ACDSA international conference.

Onboarding OS | Enterprise Multi-Tenant Lifecycle Platform

GitHub

React 19, TypeScript, Node.js, Express, PostgreSQL, Supabase, Redis, Zod, AES-256-GCM, DocuSeal

- Engineered a multi-tenant SaaS platform automating employee onboarding workflows connecting HR, IT, and managers across municipal subsidiaries (Grupo CBI).
- Implemented strict tenant isolation via PostgreSQL Row-Level Security (RLS) and transparent AES-256-GCM encryption at rest for sensitive documents.
- Built automated daily background workers for LGPD data lifecycle compliance (anonymization upon retention expiry) and state-machine-validated document pipelines.

FundiVR | Industrial Digital Twin & VR Training Simulator

GitHub

React 18, TypeScript, Fastify, PostgreSQL, Prisma ORM, WebSockets, Docker, Vitest

- Architected an industrial digital twin training simulator for aluminum furnace operators in 700°C+ environments, featuring operational safety state machines.
- Engineered real-time bi-directional telemetry via WebSockets in a modular pnpm monorepo with an automated scoring engine evaluating SOP compliance.
- Built a decoupled zero-dependency demo mode enabling standalone presentation without external database connectivity.

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura) 06/2026 – Present

Software Consultant & Solutions Architect | São Paulo, Brazil (Remote) (10/2026 – Present)

- Act as software consultant and solutions architect leading enterprise software engineering, platform development, and autonomous AI systems for mission-critical operations.
- Architect and manage cloud infrastructure on Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager) and relational databases, directly responsible for safeguarding sensitive citizen data across metropolitan public concessions in Brazil.
- Direct technical engagements and architecture alignment with enterprise platform vendors (including Google Cloud and Microsoft), ensuring rigorous LGPD compliance, zero-trust access, and infrastructure resilience.
- Conceived, architected, and deployed "Isa", an enterprise AI agent utilizing Google Gemini, autonomous tool execution via MCP Servers, contextual RAG retrieval, and deterministic JSON Schema validation with Zod.
- Conduct corporate technical enablement workshops and establish AI governance frameworks, training engineering and operational teams on secure LLM workflows.

IT Systems Analyst | São Paulo, Brazil (06/2026 – 10/2026)

- Managed enterprise identity governance, access security, and endpoint fleet telemetry across Microsoft 365, Entra ID, and Microsoft Intune.
- Engineered automated security and compliance monitoring pipelines using Microsoft Graph API and Intune, advancing overall infrastructure posture to 89% and IAM posture to 91%.
- Developed custom automation scripts in Python and PowerShell to optimize cross-departmental operations and incident deflection, resulting in promotion to Software Consultant.

Farmamed Drogarias 10/2025 – 03/2026

Full-Stack Developer & Automation | Sorocaba, Brazil

- Developed Python microservices and automation scripts for multi-site operational data consolidation, log analytics, and systems synchronization.
- Built automated pipelines for enterprise identity provisioning and endpoint governance across Entra ID, Microsoft Intune, and Active Directory.
- Designed fault-tolerant monitoring workflows and deterministic automation to resolve incidents and maintain high-availability SLAs.

Wobben Windpower 11/2023 – 04/2025

Software & IT Systems Intern | Sorocaba, Brazil

- Selected for the global Microsoft Copilot Flight Crew program, evaluating generative AI enterprise features and real-world adoption workflows.
- Developed internal automation scripts and integration routines for SAP ERP, Active Directory, and Microsoft 365 in a heavy industrial manufacturing environment.
- Provided tier-2 technical resolution for corporate service desk incidents via ServiceNow under strict SLA benchmarks.

EDUCATION

Higher Technology Degree in Systems Analysis and Development (TADS) 2026

Centro Universitário SENAC, Brazil

TECHNOLOGY RESIDENCY

ICT Resident Researcher (Digital Twins & 5G) 10/2026 – Present

UniFacens | Softex | Ministry of Science, Technology and Innovation (MCTI) | ICT Technological Residency (TIC 51)

- Selected among the top 20 out of 300 participants from the ICT Specialization (360h) to join the Technological Residency (6-month track), building industrial digital twins, real-time telemetry, and private 5G networks.
- Architected and built FundiVR, an industrial digital twin training simulator featuring real-time bi-directional telemetry via WebSockets and operational safety state machines, defended before an academic and industrial jury.

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader | Certified AppSec Practitioner (CAP) | CCNA 1: Introduction to Networks (Cisco) | Network Security & Endpoint Security (Cisco)

Languages: English (C1+ / Fluent) | Spanish (Intermediate / Technical Communication) | Portuguese (Native)

Volunteering: Volunteer English Teacher at Clube Poliglota Sorocaba, mentoring international students across cross-cultural settings.